



NPWS An tSeirbhís Páircanna
Náisiúnta agus Fiadhúla
National Parks and Wildlife Service

INSTALLING TIMBER DAMS

A STEP-BY-STEP FIELD INSTALLATION & QUALITY GUIDE



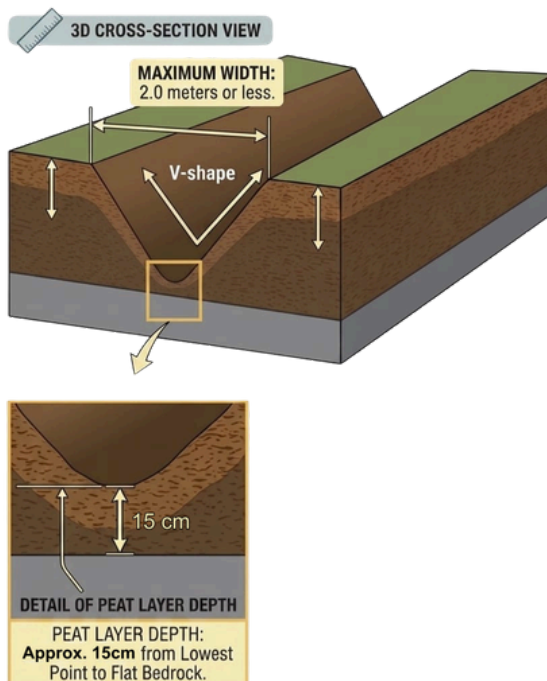
Páirc Náisiúnta Sléibhte Chill Mhantáin
Wicklow Mountains National Park

Timber dams are leaky structures designed to slow water flow and prevent gully erosion in damaged and degraded peatlands. Correct site placement and strict construction tolerances are vital for achieving overall restoration and conservation goals, and to prevent structural failure.

1. Site & Location Assessment

Ideal Topography Criteria

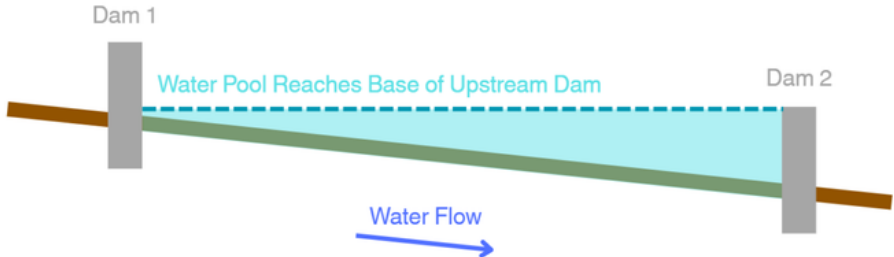
- **Gully Geometry:** The gully should feature a distinct V-shape.
- **Maximum Width:** 2.0 meters or less at its widest point.
- **Slope Gradient:** A slope angle of 6° or less.
- **Substrate Depth:** Minimum of 15cm of peat between the gully floor and the underlying bedrock to ensure secure anchoring.



Sequence & Spacing Logistics

- **Installation Flow:** Locate the first dam at the highest suitable point in the gully, and proceed progressively down the slope.
- **Catchment Spacing Rule:** The distance between individual dams is dictated by the slope. When a dam fills, the pool it creates should extend back to at least the base of the next uphill dam.
- **Height Restriction:** Never build a dam so high that its pool completely submerges the next uphill dam.

FIGURE 2: SIDE PROFILE & CORRECT SPACING SEQUENCE



2. Construction Phase Steps

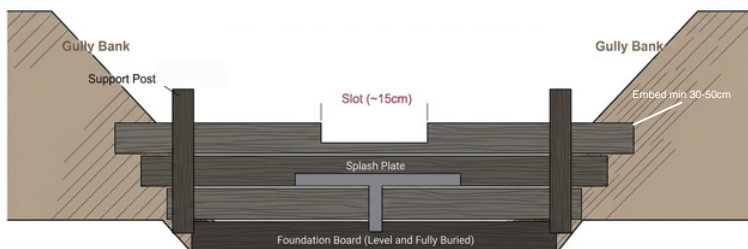
Step 1: The Foundation Board

- **Undercut Prevention:** The foundation board must be completely buried within the peat floor of the gully to minimise the risk of water undercutting the dam.
- **Bank Embedding:** The board length must be at least 60cm wider than the gully base. This allows 30cm of the board to be driven into the bank on both sides.
- **Leveling:** Ensure this board is level. The angle of the foundation board sets the geometric trajectory for all subsequent boards.

Step 2: Support Stakes / Posts

- **Orientation:** Install two support posts on the downstream side of the foundation board (wider dams may require additional posts for support).
- **Anchoring:** Drive posts deeply into the gully banks. The bank substrate is typically drier, denser, and more solid than the saturated gully bed. However, do not place them so far into the bank that it is difficult to nail the boards to the post.
- **Clearance & Spacing:** Space the posts far enough apart to ensure sufficient room to place the central flow slot and splash plate.

FIGURE 3: FRONT ELEVATION (LOOKING UPSTREAM) - STRUCTURE AND COMPONENT LAYOUT



Step 3: Subsequent Boards

- **Bank Penetration:** Embed each board added to the dam by a minimum of 30–50cm into each side bank. On wetter sites, and especially when the banks lack vegetation, 50cm is preferred.
- **Level Tolerances:** Always ensure that each board is completely level before you add another board.
- **Permeability Control:** Fit boards tightly together. While timber dams are, technically, 'leaky' structures, gaps that are too wide will allow water to escape uncontrollably.

Step 4: The Flow Slot & Splash Plate

- **The Central Slot:** Cut a slot approximately 15cm wide and 4cm deep in the centre of the top board. Wider slots are better than narrow ones, as this prevents clogging by debris.
- **Splash Plate Alignment:** Position a secure splash plate below the slot. The splash plate must be wider than the overflow slot to ensure that all spilling water is captured.
- **Height Tuning:** Avoid placing the splash plate too high (creates waterfall erosion) or too low (water overshoots and scours the gully bed).

Tools List

- ☐ Handsaw
- ☐ Claw Hammer
- ☐ Spirit Level
- ☐ Sledge Hammer
- ☐ Nails (galvanised; at least 4 inch/100mm)
- ☐ Battery Drill (optional - generally useful for splash plate; should not be used to secure boards to posts)
- ☐ Screws (if using a drill; 2–4 inch)
- ☐ Spade/toughened saw (for cutting through vegetation and root structure to install boards)

Checklist

- ☐ Gully Suitability: Gully is V-shaped, $\leq 2\text{m}$ wide, slope is $\leq 6^\circ$, and base has $\geq 15\text{cm}$ peat over bedrock.
- ☐ Sequence Verification: Construction starts at the highest suitable elevation point in the gully.
- ☐ Uphill Spacing Check: Pool level will reach the base of the uphill dam without submerging it.
- ☐ Foundation Sealing: Bottom board is fully buried into the gully bed to prevent undercutting.
- ☐ Bank Extension: Every board extends at least 30cm into both the left and right banks.
- ☐ Level Verification: The foundation and all subsequent boards have been checked with a level.
- ☐ Post Anchoring: Two downstream posts are anchored firmly into the solid, drier side-banks.
- ☐ Board Joints: Planks are fitted tightly together to allow controlled seepage without major gaps.
- ☐ Slot Geometry: Overflow slot is cut $\sim 15\text{cm}$ wide in the middle of the top board.
- ☐ Splash Plate Calibration: Plate is wider than the slot and set at an optimal height to catch spill water.



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